Problem – Basic Array Processing

1. You are to create a function that reads in a list of 10 student first names, last names and GPA into and array.

   Once completed, call another function to display the arrays.

   Create a third function to display the arrays in reverse order.

Data file

JOHN JONES 3.50

BETTY CROCKER 4.0

SAM SMITH 3.5

JOE DOE 2.75

MIKE MORE 3.0

JAVIER BAEZ 3.35

ANTHONY RIZZO 4.0

DAVID ROSS 3.90

JOE MADDON 2.80

THEO EPSTEIN 4.0

<table>
<tr><th>Input</th><th>Process</th><th>Output</th></tr>
<tr><td rowspan="2">• Student’s First Name<br>• Student’s Last Name<br>• GPA</td><td>load_arrays(Student’s First Name, Student’s Last Name, & GPA)<br>• Declare i to count loops<br>• Open Students.txt<br>• Loop that starts at 0, continues up to 9, and updates by +1<br>• Get [i]Student’s First Name, [i]Student’s Last Name, & [i]GPA<br>• End Loop<br>• Close Students.txt</td><td rowspan="2">• Student’s First Name<br>• Student’s Last Name<br>• GPA</td></tr>
<tr><td>display_arrays(Student’s First Name, Student’s Last Name, & GPA)<br>• Declare i to count loops<br>• Loop that starts at 0, continues up to 9, and updates by +1<br>• Display [i]Student’s First Name, [i]Student’s Last Name, & [i]GPA<br>• End Loop</td></tr>
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| | | |
|---|---|---|
| | *display_revarrays(*Student's First Name, Student's Last Name, & GPA)<br>• Declare i to count loops<br>• **Loop** that starts at 9, continues until 0, and updates by -1<br>• Display [i]Student's First Name, [i]Student's Last Name, & [i]GPA<br>• **End Loop** | |
| | *Main*()<br>• Declare [10]Student's First Name, [10]Student's Last Name, & [10]GPA<br>• Call *load_arrays*<br>• Call *display_arrays*<br>• Call *display_revarrays* | |

2. Read the data file below into arrays using a function. Use another function to display the arrays. Add a loop to the program that asks for a city and then displays the city and population for the value entered. If city not found, display a not found message. See below. Use a sequential search for the city.

Example:

Enter city name, ctl+z to stop:  Chicago

Chicago has a population of 4000000

Enter city name, ctl_z to stop: Denver

Denver has a population of 2500000

Enter city name, ctl_z to stop: Rockford

Rockford not found

Enter city name, ctl_z to stop: ctl+z

Goodbye. Have a nice day.

Data File

Chicago 4000000

Denver 2500000

Milwaukee 3000000

Detroit 2200000

Oklahoma 1250000

Dallas 2100000

Houston 1750000

Indianapolis 1400000

| Input | Process | Output |
|---|---|---|
| • City<br>• Population<br>• Searched City | *Load_arrays(City & Population)*<br>• Declare i to count loops<br>• Open City.txt<br>• **Loop** that starts at 0, continues up to 7, and updates by +1<br>• Get [i]City & [i]Population<br>• **End Loop**<br>• Close City.txt | • City<br>• Population |
| | *Display_arrays(City & Population)*<br>• Declare i to count loops<br>• **Loop** that starts at 0, continues up to 7, and updates by +1<br>• Display [i]City & [i]Population<br>• **End Loop** | |
| | *Search_arrays(City, Population, & Searched City)*<br>• Declare i to count loops<br>• Declare Found City to determine if Searched City was found<br>• **Loop** that starts at 0, continues up to 7 and while Found City is false, and updates by +1<br>• If Searched City equals [i]City{ Display [i]City & [i]Population<br>• If Found City is False, tell user that city couldn't be found<br>• **End Loop** | |
| | *Main()*<br>• Declare [8]City, [8]Population, & Searched City<br>• <u>Call *Load_arrays*</u><br>• <u>Call *Display_arrays*</u><br>• Assign a variable to the input for Searched City<br>• **While not EOF**<br>• <u>Call *Search_arrays*</u><br>• Ask for new entry for Searched City<br>• **End Loop** | |

3. Make an array of 10 employees (first name, last name, salary). Read the data file below and load it into the arrays. Use a function to display the arrays. Repeatedly ask the use for a last name.

Use a function to perform a sequential search on the array last name. If the last name is found display first name, last name and salary. If the last name is not found. Display last name and message “not found”.

Empldata.txt

John Jones 55000.00

Frank Smith 75000.00

Mark Smith 45000.00

Dave Roberts 80000.00

Sally Davis 90000.00

Tom Hanks 100000.00

Betty Crocker 98000.00

Sue Klein 78000.00

Joe Maddon 65000.00

Kris Bryant 70000.00

| Input | Process | Output |
|---|---|---|
| • Employee’s First Name<br>• Employee’s Last Name<br>• Employees’ Salary<br>• Search Last Name | *Load_arrays(Employee’s First Name, Employee’s Last Name, & Employees’ Salary)*<br>• Declare i to count loops<br>• Open Empldata.txt<br>• **Loop** that starts at 0, continues up to 9, and updates by +1<br>• Get [i]Employee’s First Name, [i]Employee’s Last Name, & [i]Employees’ Salary<br>• **End Loop**<br>• Close Empldata.txt | • Employee’s First Name<br>• Employee’s Last Name<br>• Employees’ Salary |
| | *Display_arrays(Employee’s First Name, Employee’s Last Name, & Employees’ Salary)*<br>• Declare i to count loops<br>• **Loop** that starts at 0, continues up to 9, and updates by +1<br>• Display [i]Employee’s First Name, [i]Employee’s Last Name, & [i]Employees’ Salary<br>• **End Loop** | |
| | *Search_arrays*(*Employee’s First Name, Employee’s Last Name, Employees’ Salary, & Search Last Name*)<br>• Declare i to count loops | |

| | | |
|---|---|---|
| | • Declare Found Employee to determine if Search Last Name was found<br>• **Loop** that starts at 0, continues up to 9 and while Found Employee is false, and updates by +1<br>• If Search Last Name equals [i]Employee's Last Name{ Display [i]Employee's First Name, [i]Employee's Last Name, & [i]Employees' Salary<br>• If Found Employee is false, tell user that Employee couldn't be found<br>• **End Loop** | |
| | *Main()*<br>• Declare [10]Employee's First Name, [10]Employee's Last Name, [10]Employees' Salary, & Search Last Name<br>• Call *Load_arrays*<br>• Call *Display_arrays*<br>• Assign a variable to the input for Search Last Name<br>• **While not EOF**<br>• Call *Search_arrays*<br>• Ask for new entry for Search Last Name<br>• **End Loop** | |